

Rethinking Evaluation for Data-Driven Flood Early Warning: Evidence from a 51-Node River Network in Sri Lanka

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Abstract—Data-driven flood early-warning systems are typically reported with a single threshold-free ranking metric, and increasingly reach high scores. We show, on a purpose-built national dataset for Sri Lanka, that such scores can be almost entirely uninformative about operational skill. We release a leakage-controlled daily panel of 51 gauged river nodes across 16 basins (2003–2024; 409,350 supervised node-days; 1,469 flood episodes) with a directed river graph, and evaluate a multimodal transformer (MMF-Net) against four baselines under three block protocols. Three findings follow. First, the standard target — next-day discharge above a per-node 98th percentile — is close to self-answering: a single-feature discharge-percentile rule attains PR-AUC 0.8164 and gradient-boosted trees 0.8496, above our best network’s 0.8355. Second, replacing the block-temporal split with a random split raises PR-AUC from 0.704 to 0.905 while pre-onset event detection *collapses* from 0.380 to 0.089: leakage does not merely inflate the score, it selects a different model that never warns before onset. Third, focal loss — near universal in imbalanced remote-sensing work — is a defect rather than a trade-off here, costing 0.0509 PR-AUC and tripling calibration error at identical architecture and seeds. We also report two independent null results for a Sentinel-1 SAR branch. We argue that flood-forecasting benchmarks should report event-level detection at a capped false-alarm ratio together with calibration, and we release the dataset, code and evaluation harness.

Index Terms—flood early warning, evaluation protocol, data leakage, calibration, tabular deep learning, graph neural networks, Sri Lanka

I. INTRODUCTION

Sri Lanka’s exposure to riverine flooding is severe and increasing. Over 1975–2025, cyclonic disturbance frequency rose from 1.3 to 4.1 systems per year (+216%, Sen’s slope +0.73/decade, $p < 0.001$), extreme rainfall intensity increased by 12.2 mm/decade, and major flood events increased 4.5-fold [1]. Cyclone Ditwah (November 2025) caused 620+ deaths and displaced approximately 1.1 million people across 21 districts — the deadliest cyclone-related event since the 2004 tsunami [1].

Critically, the same study reports that forecast lead time has already improved roughly six-fold (from 12–24 h to 120–168 h) while evacuation completion times have remained static at 18–36 h, and that the correlation between forecast accuracy and reduced casualties is weak ($R^2 < 0.4$) [1]. The binding

constraint is therefore not raw lead time. It is the translation of a forecast into action — a chain that breaks when warnings are generic, uncalibrated, or issued so often that they are discounted.

This reframes what a useful flood model must deliver. It is not a higher ranking score. It is a *calibrated probability*, at a *controlled false-alarm rate*, *before* an episode begins. Our central claim is that the metrics in common use do not measure any of those three things, and that this is demonstrable rather than rhetorical.

Contributions.

- 1) A public, leakage-controlled spatiotemporal dataset for Sri Lanka: 51 gauged nodes, 16 basins, daily 2003–2024, 409,350 supervised node-days, 1,469 labelled flood episodes, with a directed river-flow graph and three precomputed block protocols (Section III).
- 2) Evidence that the conventional next-day threshold target is dominated by discharge persistence, so that PR-AUC on it is a ceiling rather than a contest (Section V-A).
- 3) A measured demonstration that random splitting selects a model that *cannot* warn before onset, with event detection falling by 77% while the headline score rises 29% (Section V-D).
- 4) A clean negative result for focal loss, and two independent null results for a Sentinel-1 SAR branch (Sections V-B, V-G).
- 5) An evaluation protocol — event-level detection at matched false-alarm ratio, with calibration reported alongside — together with an open harness that implements it (Section III-C).

II. RELATED WORK

A. Deep learning for flood and streamflow forecasting

LSTM rainfall–runoff modelling established that a single deep model trained across many catchments can match or exceed calibrated conceptual hydrology [2], [3]. Nearing *et al.* extended this to ungauged basins globally, reporting five-day-lead reliability comparable to or better than same-day nowcasts from the Copernicus Global Flood Awareness System [4].

That result is directly relevant to us: our supervision signal is derived from GloFAS reanalysis discharge [5], a product their system outperforms. We therefore position this work not as a competitor to a global operational forecast, but as a national, node-level, fully reproducible study of *how such systems should be evaluated*.

B. Graph neural networks on river networks

Spatiotemporal GNNs have been applied to flood prediction using catchment graphs [7]–[9]. The evidence that river topology helps is contested. Kirschstein and Sun report that their model “fails to benefit from the river network topology information, both on the entire network and small subgraphs,” with learned edge weights uncorrelated with static topology [10]. Wang *et al.* reach the opposite conclusion, but only after a reachability-based graph transformation that mitigates over-squashing [11]. Our own graph-based family (Section V-B) is consistent with the former; we treat the question as open and report it as such.

C. Tabular deep learning

Gradient-boosted trees remain strong on medium-sized tabular data [12]. The published remedy is not depth but the input layer: per-feature periodic embeddings substantially close the gap [13]. Our single largest gain reproduces this on a hydrological panel (Section V-B).

D. Evaluation, leakage and calibration

Roberts *et al.* specify block cross-validation for temporally and spatially structured data [14]. Kapoor and Narayanan document leakage as a reproducibility crisis across 17 fields, with a taxonomy of eight leakage types [15]. Precision–recall analysis is preferred to ROC under strong class imbalance [16], [17], and POD/FAR/CSI are the standard meteorological verification quantities [18]. Modern networks are badly calibrated by default; temperature scaling is the standard post-hoc remedy [19], and deep ensembles provide both accuracy and uncertainty [20].

E. Sri Lankan flood prediction

Prior Sri Lankan work is concentrated on single stations or single districts, and — crucially for this paper — reports metrics that conceal the failure mode. Table I summarises what the published numbers rest on.

Saubhagya *et al.* forecast water level at Ratnapura (Kalu basin) with a Bi-LSTM plus a GLS regression and ARIMA residual model, using six predictors over 2014–2019 [24]. Their protocol is sound — an expanding-window temporal split — and is the methodologically strongest local work. However, their headline “accuracies of 80%, 80% and 100%” for actionable events are computed over the three Minor and two Critical events that occur in the whole 2017–2019 evaluation window; the Major class never occurs at all. Their reported overall accuracy also sits at or below the majority-class rate in every year (e.g. 92% reported for 2019 against a 98.6% “never predict a flood” rate). They state explicitly that

TABLE I
SCALE AND EVIDENCE BASE OF PRIOR SRI LANKAN FLOOD-PREDICTION STUDIES, COUNTED FROM EACH PAPER’S OWN TABLES.

Study	Spatial units	Period	Resol.	Events eval.
[24]	1 target + 4 upstream	2014–19	daily	5
[25]	1 district	1976–2015	monthly	51 records
[26]	6 catchments	n/s	n/s	n/s
This work	51 nodes / 16 basins	2003–24	daily	1,469

“a better performance could have been achieved if data from a higher number of nearby WL gauging stations were available” [24] — the gap a national node network addresses.

Thilakarathne and Premachandra predict monthly flood occurrence for Anuradhapura using ARIMA-forecast weather and an ANN classifier, reporting 91.7% accuracy under 10-fold cross-validation [25]. Their own table reports mean recall 0.404, with two of ten folds at precision = recall = 0.000. Ten-fold cross-validation on a monthly time series is a random split, i.e. exactly the protocol characterised as invalid in [14], [15].

Mahaganapathy *et al.* report a 99% F1 score for Kalu Ganga catchments using SMOTE and ensemble classifiers [26]; the available abstract does not state the splitting protocol.

Operationally, the Department of Irrigation issues basin warnings from observed rainfall, catchment soil moisture, river water level and river trend [28] — a feature set our 33 dynamic channels deliberately mirror. Official Kelani thresholds at the Nagalagam Street gauge are 5.0–7.0 ft (minor), > 7.0 ft (major) and > 9.0 ft (dangerous) [27].

III. DATASET AND PROBLEM FORMULATION

A. Node network and graph

We define 51 river nodes across 16 basins, each tagged with a climatic zone (wet / intermediate / dry) and a network position (upstream / mid / downstream / outlet). Fig. 1 shows the network. The graph carries two *separately typed* relations that are never symmetrised: 35 directed upstream→downstream *flow* edges curated from a `downstream_of` field, and 204 distance-decayed bidirectional *spatial* *k*-NN edges, plus self-loops. Edge attributes are $[w, d_{\text{km}}/100, \Delta z/100, \mathbf{1}_{\text{flow}}]$, where Δz is the elevation drop from source to destination.

B. Features and targets

Each node-day carries 33 dynamic channels — precipitation and multi-day accumulations, an antecedent precipitation index, temperature, humidity, wind, radiation, three soil-wetness depths with anomalies, and eight discharge-derived channels — plus 9 static terrain features. Heavy-tailed channels are $\log(1 + x)$ transformed before standardisation. Sources are NASA POWER (meteorology), GloFAS via Open-Meteo (discharge) [5], ERA5-Land [6] and Copernicus DEM.

The primary target is

$$y_{n,t} = \mathbf{1}[Q_{n,t+1} > \tau_n], \quad \tau_n = P_{98}(Q_{n,\cdot}^{\text{train}}), \quad (1)$$

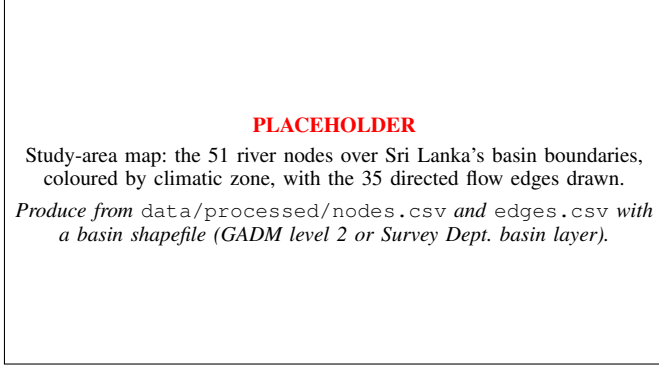


Fig. 1. The 51-node river network. Node identity, basin, zone and position are given in `scripts/common/nodes.py`.

TABLE II
TEMPORAL PROTOCOL BLOCK SIZES (VALID SUPERVISED NODE-DAYS).

Block	Dates	N	Positive rate
Train	2003-01-01 – 2017-12-31	277,899	2.028%
Val	2018-01-01 – 2020-12-31	55,896	0.810%
Test	2021-01-01 – 2024-12-31	75,555	2.275%
Total		409,350	

i.e. next-day discharge at node n exceeding that node’s own 98th-percentile threshold, with τ_n fitted on the training block only. Three auxiliary classification heads (2-day, 3-day, and *onset*, defined as a flood within the horizon while not currently flooding) and two regression heads are trained jointly. A `label_confidence` weight in $[0.5, 1]$ down-weights days within $\pm 15\%$ of τ_n .

C. Protocols

Random splitting is structurally impossible in this release: three block protocols ship precomputed. The *temporal* protocol trains on $\leq 2017-12-31$, validates on 2018–2020 and tests on 2021 onward; the *basin* protocol holds out the entire Gin basin; the *event* protocol groups by episode identifier. A diagnostic random split exists solely to *measure* leakage (Section V-D). All normalisation statistics, percentile thresholds and day-of-year climatologies are fitted on the training block alone. Table II gives the sizes.

The validation block is hydrologically quieter than train and test (0.810% vs 2.028%/2.275%). This is real hydrology, not a sampling artefact, and it makes early stopping on node-day PR-AUC an optimistic-biased signal; we therefore select on an episode-level criterion instead (Section IV-C).

IV. METHOD

A. MMF-Net

Fig. 2 shows the architecture. A batch is a set of *days*; every forward pass sees the whole network, so that spatial variants are possible without changing the data path.

Given a window $\mathbf{x} \in \mathbb{R}^{B \times L \times N \times F}$ ($L = 14$ days, $N = 51$ nodes, $F = 33$ channels) the nodes are folded into the

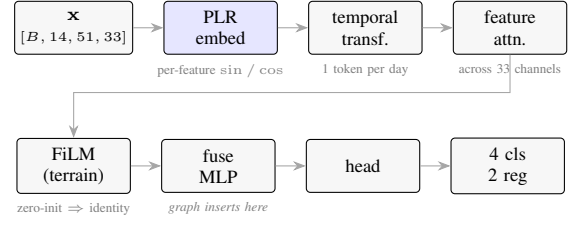


Fig. 2. MMF-Net. Nodes are folded into the batch dimension, so the encoder is per-node; the graph-based family of Section V-B inserts relational message passing between *fuse* and *head*. Parameter shares for the 711k headline model: temporal transformer 61.7%, feature attention 19.4%, head 9.3%, fusion 7.0%, FiLM 2.5%.

batch dimension and each scalar is given its own *periodic-linear-ReLU* (PLR) embedding [13]: for feature j with learned frequencies $\mathbf{c}_j \in \mathbb{R}^k$,

$$\mathbf{e}_j(x) = \text{ReLU}(\mathbf{W}_j [\sin(2\pi \mathbf{c}_j x) \parallel \cos(2\pi \mathbf{c}_j x)] + \mathbf{b}_j). \quad (2)$$

The motivation is precise: a linear layer applied to a raw scalar is monotone and cannot represent a sharp decision boundary, which is exactly what a decision tree obtains for free from a split point.

Per-day embeddings are concatenated, projected to one token per day, and pooled by a CLS token through a three-layer pre-LN transformer. A second, cheaper attention block operates *across* the 33 channels of the time-averaged window, expressing conditional interactions (“this much rain matters only when the soil is already wet”) that a single linear mixing layer cannot. Static terrain then modulates the state by FiLM [22], $\mathbf{h} \leftarrow \text{LN}(\mathbf{h} \odot (1 + \gamma) + \beta)$, with the generator’s output layer zero-initialised so the module begins as exactly the identity. A fusion MLP and a shared head produce four classification logits and two regressions.

B. Training, calibration and thresholding

All heads are optimised jointly with a masked weighted sum of per-head binary cross-entropy plus a Huber term for the regressions. Optimisation is AdamW, $\text{lr} = 10^{-3}$, cosine schedule with five warmup epochs, 60 epochs with early stopping (patience 10). Five seeds are trained independently and their logits averaged [20]; a single temperature is then fitted *on validation* [19] and the decision threshold is likewise chosen on validation and frozen. The test block is touched once.

C. Metrics

We report PR-AUC [16], [17], Brier score, expected calibration error, and POD/FAR/CSI [18]. The operationally decisive quantity is *pre-onset event detection* (`ev.det`): the fraction of flood episodes for which the model raises an alarm in the seven days *before* onset, with mean lead time among detected episodes. Model selection uses an episode-level PR-AUC in which each episode contributes one positive, scored by the model’s maximum probability in the pre-onset window, and each flood-free node-week contributes one negative.

TABLE III
BASELINES, TEMPORAL PROTOCOL, TEST BLOCK.

Baseline	PR-AUC	ROC-AUC	POD	FAR	ev.det
Persistence	0.5900	0.882	0.769	0.231	0.223
Climatology	0.1080	0.870	0.556	0.882	0.515
Discharge percentile	0.8164	0.984	0.769	0.231	0.223
GBT (LightGBM)	0.8496	0.988	0.634	0.080	0.161

TABLE IV
MMF-NET ABLATION LADDER, TEMPORAL PROTOCOL, TEST BLOCK. $\times 5$ DENOTES A FIVE-SEED ENSEMBLE WITH TEMPERATURE SCALING.

Rung	Change	PR-AUC	ev.det	FAR	Params
N0	linear embeddings	0.6083	0.290	0.412	551k
N1	+ PLR embeddings	0.7605	0.420	0.325	555k
N2	+ feature attention	0.7738	0.423	0.313	693k
N3	+ FiLM terrain	0.8269	0.197	0.110	711k
N4	+ focal loss	0.7592	0.220	0.138	711k
N5 $\times 5$	+ ensemble (focal)	0.7846	0.208	0.145	711k
N5_bce$\times 5$	the same, BCE	0.8355	0.220	0.120	711k
N6_sc $\times 5$	+ SAR scalars	0.7284	0.397	0.344	719k
N6_gat $\times 5$	+ gated SAR CNN	0.8310	0.217	0.107	11,967k

V. RESULTS

A. Baselines: the target is close to self-answering

Table III gives four reference models under the temporal protocol. The decisive row is `discharge_pctl1`: a single feature, no learning, PR-AUC 0.8164. Because the target is next-day discharge above a percentile and discharge is strongly autocorrelated, today’s percentile nearly answers the question. Gradient-boosted trees add only 0.033 on top of that.

The consequence is that PR-AUC on this target measures persistence more than skill, and that the genuine headroom lies elsewhere: the best baseline warns before onset on only 22.3% of episodes, and the strongest baseline (GBT) on 16.1%. Climatology’s apparently high 0.515 is an artefact of alarming almost constantly (FAR 0.882), which is precisely the confound Section V-E addresses.

B. Ablation ladder

Table IV and Fig. 3 give the MMF-Net ladder, each rung changing exactly one thing.

The input layer, not depth, closes the gap to trees. N0→N1 changes only the numerical embedding and moves PR-AUC by +0.152 — the largest single-change gain in the study, and larger than terrain conditioning (+0.105 in our graph-based family). This reproduces [13] on a hydrological panel and supports the diagnosis of [12].

Focal loss is a defect, not a trade-off. N5→N5_bce changes *only* the loss — identical architecture, seeds, ensemble and calibrator — and gains +0.0509 PR-AUC while reducing ECE from 0.0043 to 0.0016; the single-seed comparison N3 (BCE, ECE 0.0032) against N4 (focal, ECE 0.0525) is starker still. Given the ubiquity of focal loss [21] in imbalanced geoscience pipelines, we report this as a transferable negative result.

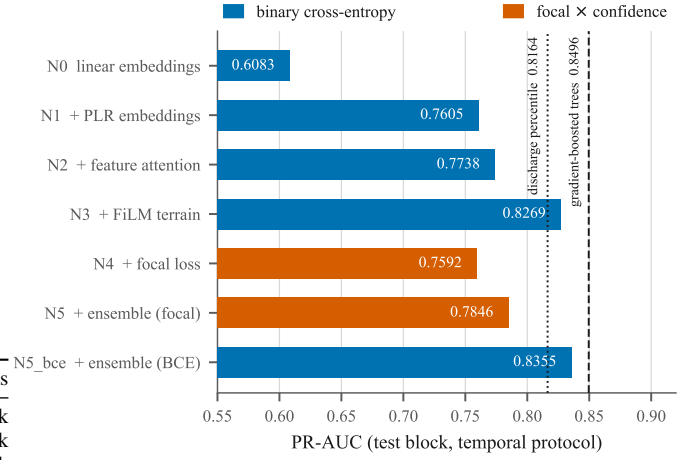


Fig. 3. PR-AUC up the ladder. The two reference lines are the discharge-percentile rule and gradient-boosted trees. Bars are coloured by loss function: the focal-loss rungs (N4, N5) form a visible regression that reverting to BCE removes.

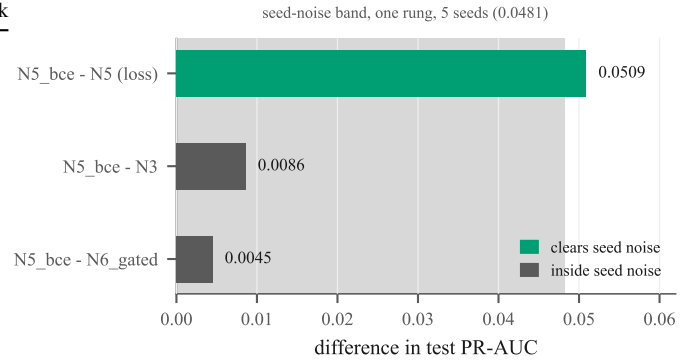


Fig. 4. Differences between configurations against the seed-to-seed spread of a single configuration. Only the loss effect is resolvable by this experiment.

The best network still loses to trees. N5_bce reaches 0.8355 against the GBT’s 0.8496 — best in the study on PR-AUC, ECE (0.0016), Brier (0.0080), CSI (0.561) and F1 (0.719), at 711k parameters, but 0.0141 short of a gradient-boosted tree and only 0.019 above a one-line percentile rule.

C. Differences at the top are inside seed noise

Across N5_bce’s five seeds, best-validation episode PR-AUC ranged 0.3375–0.3856, a spread of 0.0481. Fig. 4 places the differences being ranked against that spread: the N5_bce–N6_gated (0.0045) and N5_bce–N3 (0.0086) gaps are well inside it, whereas the loss effect (0.0509) clears it. We therefore claim the loss result and explicitly *decline* to claim an ordering among the top three configurations.

D. Leakage selects a different model

Table V and Fig. 5 report one architecture under three protocols, with every hyperparameter held fixed.

Under a random split PR-AUC rises from 0.704 to 0.905 (+29% relative), above the GBT baseline and above every honest configuration we trained. Its pre-onset event detection

TABLE V
ONE MODEL, THREE PROTOCOLS. THE RANDOM SPLIT IS DIAGNOSTIC ONLY.

Protocol	PR-AUC	Brier	POD	FAR	ev. det
Block temporal	0.704	0.0135	0.621	0.342	0.380
Basin holdout	0.840	0.0078	0.498	0.081	0.217
Random	0.905	0.0051	0.753	0.113	0.089

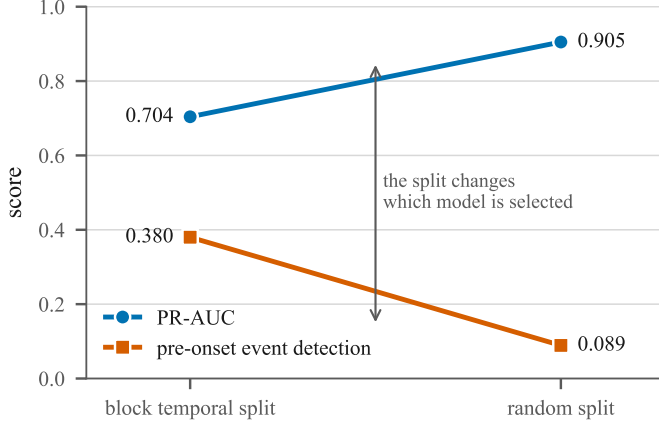


Fig. 5. The same architecture under a block-temporal and a random split. PR-AUC rises 29% relative while pre-onset event detection falls 77%. A study reporting only the former would conclude the random-split model is far better; it is in fact operationally useless.

simultaneously *collapses* from 0.380 to 0.089. The random-split model has learned to interpolate between neighbouring days — days that are, under that protocol, in the training set — and has stopped forecasting altogether.

That PR-AUC is inflated by leakage is known [14], [15]. The contribution here is the second half: leakage does not merely add a bias to a score, it *changes which model is selected*, in the direction that destroys operational value. Any pipeline selecting on a leaked ranking metric will systematically prefer the model that cannot warn.

We note that the basin-holdout row is *not* comparable to the temporal row: that protocol still trains on the full 2003–2024 span and so retains temporal leakage by construction. It measures spatial transfer only, and its 0.840 should never be quoted as this study’s best result.

E. Operating points, not models

Fig. 6 plots detection against false-alarm ratio for every configuration. FAR spans 0.080 to 0.882. Reading *ev. det* down a results column therefore compares thresholds rather than models: N2’s 0.423 at FAR 0.313 and N3’s 0.197 at FAR 0.110 are not a regression in skill but a move along an operating curve, and N3’s threshold-free PR-AUC in fact *rose*. We supply an offline utility that re-derives every threshold to a common false-alarm budget, and we recommend that flood papers report detection at a stated FAR rather than at each model’s own optimum.

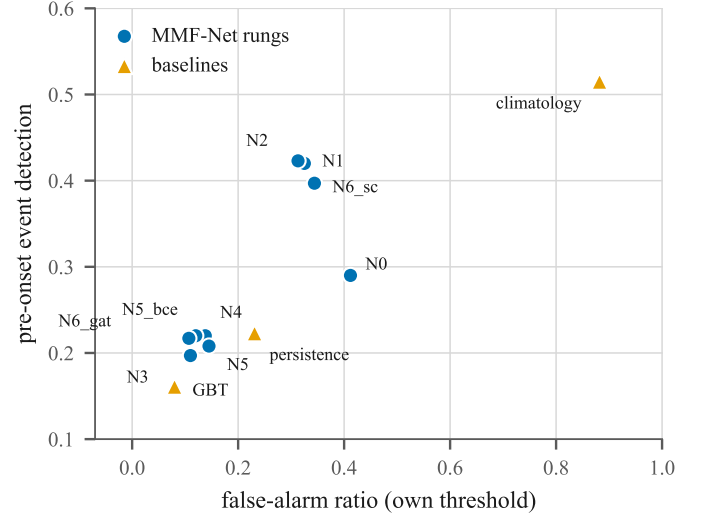


Fig. 6. Pre-onset event detection against false-alarm ratio at each model’s own validation-chosen threshold. The spread in FAR is why the detection column of a results table is not comparable down the page.

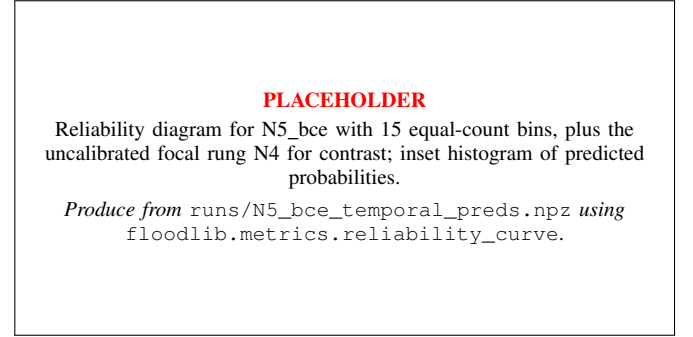


Fig. 7. Calibration of the headline model after temperature scaling.

F. Calibration

The ensemble plus temperature scaling reduces expected calibration error to 0.0016 for N5_bce, against 0.0525 for the uncalibrated focal-loss rung. Fig. 7 gives the reliability diagram.

G. The SAR branch does not help

A Sentinel-1 branch was evaluated twice. In our graph-based family a ResNet-18 embedding concatenated into the fusion input was null. In MMF-Net, a pretrained encoder fused through a learned gate (zero-initialised with a negative bias, so the module begins closed) reached 0.8310 with 11,967k parameters — *below* the 711k image-free model. What had appeared to be a vision gain was the BCE loss recovering what focal loss had destroyed.

We report this as a *coverage* result rather than a claim about SAR for flood mapping in general [23]: imagery reaches only 9 of 51 nodes, and Sentinel-1’s ~12-day revisit is structurally mismatched to a one-day horizon. Lead time in this system must come from the daily meteorological and discharge channels.

VI. DISCUSSION

Three practical recommendations follow.

Report event-level detection at a capped false-alarm ratio. A threshold-free ranking metric on a next-day threshold target is dominated by persistence, and a detection rate at each model’s own optimum is not comparable across models. Neither supports an operational decision.

Report calibration alongside. The evidence from [1] is that the binding constraint in Sri Lanka is warning-to-action, and that over-frequent or discounted warnings break that chain. A probability that is right on average at every confidence level is therefore not a nicety.

Treat the split as part of the model. Our random-split configuration would, on the usual headline metric, be reported as the best model in this paper. It is the worst. Because a leaked protocol changes model *selection*, no amount of downstream metric care compensates for it.

VII. LIMITATIONS

No embargo between blocks. The 14-day lookback is indexed absolutely, so test days in the first 13 days of 2021 read features from December 2020, which lies in the validation block. This is causal and not label leakage, but it is uncontrolled and a 13-day embargo would remove it.

Same nodes in train and test. The temporal protocol evaluates the same 51 gauges at later dates. Spatial transfer is measured only by the Gin-basin holdout, on 3 nodes, and in our graph-based family alone.

Labels are model-derived. Supervision comes from GloFAS reanalysis discharge [5], not from gauge observations or inundation records; episodes were cross-checked against documented events but the label is a reanalysis product with its own biases. Nearing *et al.* report outperforming GloFAS [4], which bounds what a model fitted to it can mean.

Single-seed rungs. N0–N4 are single seeds; given the measured spread of 0.0481, individual rung-to-rung differences below roughly 0.02 PR-AUC should not be read as orderings.

RQ on river topology unresolved. Whether directed flow edges help is not settled by this study, consistent with the disagreement between [10] and [11]. An encoder-matched comparison is in progress.

VIII. CONCLUSION

On a national, leakage-controlled dataset for Sri Lanka we find that the metric conventions of data-driven flood forecasting can be actively misleading: the usual target is dominated by discharge persistence, the usual splitting protocol selects models that never warn before onset, and a widely used imbalance loss is a net regression. Our best network is genuinely well calibrated and compact but does not beat gradient-boosted trees on the headline score — and we argue that this is the correct thing to report, because the headline score is not the thing that matters. We release the dataset, the models and an evaluation harness that implements event-level detection at matched false-alarm ratio, so that subsequent work on this basin network is comparable by construction.

REPRODUCIBILITY

Dataset, model code, preset ladders and evaluation harness: [repository URL]. All results in this paper are produced by `models/kaggle_run.py` and rendered by `models/results_doc.py`; figures are generated by `paper/make_figures.py`.

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